

Effective Quantum and Gravitational Structure from the Compact Fiber: Cross-Frame Projection, Hilbert Representation, Born Rule, and Coordinate-Time Metric

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Abstract

The compact fiber $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ and its degree-2 covering map $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$, derived in Paper I from five postulates of discrete scalar motion, support both a complete effective quantum-mechanical formalism and an effective gravitational metric as representational layers rather than as primitive ontology.

This paper derives the structures that standard quantum mechanics treats as axiomatic through explicit lemma chains with staged dependency tracking: complex amplitudes from the vibrational quarter-turn on the covering fiber (Proposition 4.2), the inner product from bilinear integration, weight conservation, and dynamical continuity (Lemmas 6.1–6.2), the Born rule $P = |c|^2$ from phase gauge invariance and L^2 uniqueness (Theorem 6.6), effective Hilbert space from three physical coarse-grainings each with a named domain of validity (§7), and Bell correlations $E(\theta) = -\cos 2\theta$ for photon polarization and $E(\theta) = -\cos \theta$ for spin- $\frac{1}{2}$ from covering coherence at degree 2 (Theorem 9.2). The cross-frame projection through π (Lemmas 5.1–5.5, Theorem 5.6) dissolves the measurement problem into a derived part (decoherence produces classical weights; Bayesian update follows) and an underdetermined part (single-outcome selection, structurally dissolved via projective information loss through the covering kernel).

A parallel gravitational program derives the coordinate-time effective metric from the reciprocal partition of the gravitational potential, yielding PPN parameters $\beta = \gamma = 1$ identical to general relativity, the equivalence principle as an identity of the displacement algebra (Proposition 14.1), and all four classical tests (perihelion precession, light deflection, gravitational redshift, and Shapiro time delay). Two second-order predictions distinguish the framework from GR: opposite-sign second-order redshift coefficients ($+x^2/2$ versus $-x^2/2$, testable at neutron star surfaces) and polarization-dependent light deflection (testable by VLBI).

Every claim is assigned a derivation status (§1.5); structures imported from mathematics versus those derived from the postulates are audited explicitly (§1.7). The quantum program is anchored by the Ravets et al. ^{87}Rb Rydberg Förster resonance experiment (C_3 reproduced to within 17%, R^{-3} scaling confirmed).

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1 Introduction

1.1 Aim

This paper demonstrates that effective quantum structure—the apparatus of Hilbert space, complex amplitudes, the Born rule, unitary evolution, superposition, entanglement, Bell correlations, and the resolution of the measurement problem—emerges as a representational layer on the compact fiber established in Paper I [1], rather than as primitive ontology.

The guiding principle is:

Paper I proves the compact local carrier. Paper II derives atomic structure from it. The present paper shows how effective quantum formalism and effective gravitational structure emerge as representational layers on that carrier.

All derivations in this paper rest on the five postulates and the fiber derived in Paper I; we import only the structures listed in §2.

1.2 Why the representational program is separate from the atomic paper

The original unsplit manuscript contained both local atomic structure (spectroscopic thresholds, screening quantities, multiplet ratios) and effective quantum representation (Hilbert space, Born rule, Bell correlations) in a single document. These are conceptually distinct:

The **atomic paper** (Paper II [2]) treats the local material organization of the periodic table: how the compact fiber’s displacement algebra produces spectroscopic energy levels, screening quantities, and multiplet structure.

The **present paper** treats the representational lift: how the compact fiber and cross-frame projection induce the formal apparatus of quantum mechanics as an effective description. Its content is the derivation of QM’s axioms from the fiber geometry.

Both rest on the compact fiber from Paper I, but they address different questions and their derivation chains are largely independent.

1.3 What this paper does and does not do

This paper derives effective quantum structure from the compact carrier and three physical inputs (bilinear integration, conservation of total weight, continuity of dynamics). It does not take Hilbert space, the Born rule, or wave-function collapse as primitive. It does not derive atomic screening tables, the restoration-depth law, or detailed spectroscopic benchmarks—those belong to Paper II.

1.4 Dependency on Papers I and II

From Paper I we import:

- The compact fiber $F \cong \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ with $|F| = 128$.
- The degree-2 covering map $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$.
- Sector capacities $N_m = 4$ (magnetic), $N_e = 8$ (electric).
- The translation-invariant transport metric ($1/N$ per step).

- Regime qualification: stable local material closure.

From Paper II we import:

- Time-region mechanics: the second-power relation $r = \rho^2$, the harmonic potential, and the Kustaanheimo–Stiefel bridge.
- The displacement triple (a, b, c) as fiber coordinate.
- The Class I/II quantity distinction (n_{eff} projection-invariant, Z_{eff} projection-dependent).
- The covering-modulation rule (covering acts only on fiber-read quantities).
- The five projection families as empirical validation of the carrier architecture.

1.5 Derivation status and proof obligations

This paper mixes several epistemic levels. To prevent misidentification of effective structures as primitive ones, and to make the proof burden visible, every major claim is assigned one of five status labels:

T — Theorem-grade

Complete chain from imported structure plus stated inputs. No gaps.

S — Structural consequence

Follows under explicitly named assumptions, each of which is identified.

I — Interpretive identification

Maps a derived structure onto a known physical concept. The mapping is motivated and consistent but does not constitute a derivation.

E — Empirical anchor

A regime in which the derived structure is quantitatively tested. Does not prove the abstract claim; validates that the conditions for the claim are met in a specific physical system.

O — Open formalization task

Identified gap. The claim is physically motivated but not yet proved as a separate proposition.

The headline claims of this paper:

Claim	Status	Depends on
Covering fiber is 2D real vector space	T	Paper I fiber + Prop. 4.2
Complexification via quarter-turn J	T	P2, P4, uniqueness of J (Prop. 4.2)
Inner product from L^2 uniqueness	T	Bilinearity + conservation + continuity (Lem. 6.2)
Cross-frame projection is rank-1 linear	T	π -channel uniqueness (Lem. 5.1) + displacement additivity (Lem. 5.3)
Born rule $P = c ^2$	S/T	Projection linearity + phase gauge (Thm. 6.6)
Preferred basis = coupling eigenbasis	T	Diagonal coupling + decoherence (Prop. 8.1)
Bell $E(\theta) = -\cos 2\theta$ (photon)	T	Datum neutrality $\rightarrow$ singlet (Lem. 9.1) + projection (Thm. 5.6)
Tsirelson bound from degree 2	S	Bell result + covering degree
Pauli exclusion	T	Kernel action (Lem. 5.1) + rotational rep (Prop. 4.2)
Förster C_3 and R^{-3} scaling	E	Composite selector decomposition
<i>Gravitational program</i>		
Equivalence principle	T	Identity via displacement content (Prop. 14.1)
PPN $\beta = 1$	T	Reciprocal partition of gravitational potential (§14.4)
PPN $\gamma = 1$ (1st order)	S	Reciprocal postulate on spatial metric
Four classical tests	T	Coordinate-time mechanism + $\beta = \gamma = 1$
Second-order redshift ($+x^2/2$ vs. GR $-x^2/2$)	E	Reciprocal partition; testable at neutron stars
Polarization-dependent deflection	E	Interaction cross-section; testable by VLBI

In the previous draft, two claims marked **S** depended on items with status **O**: the cross-frame factorization and the antisymmetric state selection. Both have now been upgraded:

1. The cross-frame factorization (Lemma 5.3) is promoted to **T** by recognizing that displacement additivity is a consequence of the postulates, not a modeling choice. The additive structure is testable: the quantum-defect law is additive in the displacement product and works across 95 atom-observable pairs.
2. The antisymmetric state selection (Lemma 9.1) is promoted to **T** by deriving it from P5 (datum neutrality)¹ via the total quarter-turn generator J_{total} . The singlet is the unique state annihilated by J_{total} . This is testable: the singlet predicts rotational invariance of correlations, which the triplet violates; experiments confirm rotational invariance.

The remaining **O** item is single-outcome selection (the measurement problem), which is open in all formulations of QM.

1.6 Canonical structure vs. canonical representation

Because this paper is vulnerable to the charge that it reconstructs QM using QM-shaped tools, we distinguish sharply between two things throughout:

¹P5 (datum neutrality): “The shared datum is the neutral condition relative to which all admissible structural invariants vanish. Restoration to datum requires annihilation of every invariant the closure structure preserves.”

1. **What is forced by the compact fiber** (canonical structure): the two-sheet covering, the kernel action Ω , the cyclic vibrational phase, the covering degree $d = 2$.
2. **What is a faithful representation of that structure in familiar mathematics** (canonical representation): the real 2D rotational model $V_b \cong \mathbb{R}^2$, complex amplitudes $\mathbb{C}^1$, the Hilbert-space completion after CG1–CG3.

The complex numbers, the inner product, and the Hilbert-space language are not smuggled in. They are the most economical algebraic representation of structures that the fiber forces. §1.7 provides a line-by-line audit of what is imported from mathematics versus what is derived from the postulates.

1.7 No-import audit

Structure	Imported?	Justification
2D real fiber space V_b	No	Faithful rep. of degree-2 covering (Prop. 4.2)
Complex numbers on V_b	No	Quarter-turn J on the fiber (Prop. 4.2)
Inner product on V_b	No	Bilinearity + conservation + continuity (Lem. 6.2)
Riesz representation	Math thm	Applied <i>after</i> inner product established
Gleason's theorem	Math thm	Consistency confirmation only, not used in derivation
Parseval/Plancherel	Math thms	Extend already-derived L^2 rule to continuous observables
Hurwitz's theorem	Math thm	Consistency check on $\mathbb{C}$ uniqueness at dim 2
Tensor product $\otimes$	Math construction	Applied <i>after</i> CG3 justified by spatial independence

The mathematical theorems (Riesz, Gleason, Parseval, Plancherel, Hurwitz) are tools applied to structures that have already been derived from the postulates. They do not smuggle in quantum mechanics. The derivation would still hold if these theorems were proved inline; they are cited for economy, not as hidden physical assumptions.

2 Imported Structure from the Compact Fiber

2.1 Compact carrier and covering map

The compact fiber $F \cong \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ was derived in Paper I as the unique admissibility structure for discrete scalar motion in reciprocal three-dimensional space and three-dimensional time, subject to five postulates (P1–P5). The covering map

$$\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}, \quad \pi(k) = k \bmod 4,$$

is the unique degree-2 homomorphism with kernel $\ker(\pi) = \{0, 4\} \cong \mathbb{Z}/2\mathbb{Z}$. It connects the electric (one-dimensional, temporal) sector to the magnetic (two-dimensional, spatial) sector.

The covering degree $d = N_e/N_m = 8/4 = 2$ is the fundamental structural constant of the projection. It is not a parameter; it is forced by the sector capacities, which are themselves derived from the postulates.

2.2 Stable local regime and representational scope

The present paper treats the representational layer induced by the stable local compact regime. Not every scalar-motion regime produces quantum-formal structure in the same way. The ultra-long-range regime (cosmic sector), the time-region boundary, and the transition between material and cosmic sectors are outside the scope of this paper.

2.3 What is used downstream

Four specific structures from Paper I enter the derivations below:

1. **The electric covering fiber** $F_b = \pi^{-1}(b)$, a two-element set at each magnetic base point $b \in \mathbb{Z}/4\mathbb{Z}$.
2. **The translation-invariant metric** on the fiber, providing the inner-product seed.
3. **The covering degree** $d = 2$, controlling the interference amplification.
4. **The regime distinction** between primitive structures (scalar motions, displacements) and effective structures (Hilbert space, Born rule).

3 Time-Region Representation

3.1 Particle and wave as reciprocal frame descriptions

The particle description and the wave description are not two different entities; they are two descriptions of the same scalar-motion entity, viewed from two reciprocal reference frames:

The **particle** is the entity described from the three-dimensional spatial reference frame. It has a trajectory, a definite position, and obeys Newtonian mechanics in the time-space region.

The **wave** is the same entity described from the three-dimensional temporal reference frame. It has a definite frequency, a wavefront geometry, and obeys the Schrödinger equation in the time region.

The correspondence is exact: the wave speed $u = c^2/v$ is the reciprocal of the particle speed v , expressed in the temporal frame. This follows from the reciprocal postulate—space and time are reciprocal aspects of motion—without any additional assumption. *Status: T.*

3.2 Why the wave function lives in the time region

The wave function $\psi(\rho)$ is a function of equivalent-space coordinate ρ in the time region. It does not exist in extension space. It is not a spatial object that “collapses”; it is a temporal-frame description that is *projected* onto the spatial frame through the cross-frame projection mechanism derived in §5.

The ontological status is precise: ψ describes the displacement structure of the scalar-motion entity in equivalent space. Below the natural unit of space, where the spatial component is fixed at its minimum value (P1), all variability is in the temporal component t , and the equivalent space is $1/t$.

This contrasts sharply with the standard QM treatment, in which ψ is a function on configuration space (spatial coordinates of all particles). In that framework, ψ has no clear ontological home—it is “not a wave in physical space” (for multi-particle systems) yet is treated as the complete description of reality. In the RST, ψ has a definite ontological home: it lives in the time region (equivalent space), and the apparent spatial structure emerges through the cross-frame projection. The “mystery” of the wave function’s non-spatial character is dissolved: it is non-spatial because it is temporal.

3.3 Schrödinger equation as time-region mechanics

The Schrödinger equation is the governing equation for motion in the time region. Following Nehru [4], its derivation from the energy-momentum relation for sinusoidal waves in equivalent space introduces nothing against the postulates. The time-region potential

$$V(\rho) = \frac{1}{2}\omega^2\rho^2 + \frac{K_{G1}}{\rho^2}$$

(derived in Paper II from the space-time progression and gravitation, both transformed by the second-power relation) produces the eigenvalue structure that, through the KS bridge $r = \rho^2$, maps to the observed Rydberg spectrum in extension space.

Status: The Schrödinger equation as time-region mechanics is **I** (interpretive identification following Nehru). The potential form is **T** from Paper II. The exact normalization of K_{G1} from displacement numbers to natural units is **O**.

3.4 Why this does not yet amount to primitive Hilbert space

The Schrödinger equation provides: (i) a linear differential equation with solutions in $L^2(\mathbb{R}^+)$; (ii) eigenvalues and eigenstates; (iii) time evolution via $e^{-iHt/\hbar}$. What it does *not* provide by itself: (a) why complex amplitudes are physically meaningful rather than mathematical convenience; (b) why the squared modulus $|\psi|^2$ gives probabilities; (c) what happens at measurement; (d) why the spatial basis is preferred for outcomes; (e) why entangled states exhibit non-classical correlations. These five items are what the present paper derives from the covering-fiber geometry and cross-frame projection.

4 The Covering Fiber as a Real Vector Space

This section establishes the vector-space structure of the covering fiber through a chain of three results (Definition 4.1, Proposition 4.2, and the Hurwitz consistency check), each with explicit premises and proof.

4.1 Covering fiber at a base point

Definition 4.1 (Covering fiber). *At each base point $b \in \mathbb{Z}/4\mathbb{Z}$, the covering fiber is the preimage*

$$F_b = \pi^{-1}(b) = \{b, b + 4\} \subset \mathbb{Z}/8\mathbb{Z}.$$

This is a set of $d = 2$ elements (the covering degree), corresponding to the two vibrational phases of the primitive vibration in scalar dimension 3 (postulate P4).

The two elements of F_b are physically distinct: they represent the two phases of the vibrational motion that occupies one of the three scalar dimensions. $|s_1\rangle$ corresponds to the vibration at phase 0 (one half-cycle) and $|s_2\rangle$ corresponds to the vibration at phase π (the other half-cycle). The kernel of π acts by exchanging these two phases:

$$\Omega|s_1\rangle = |s_2\rangle, \quad \Omega|s_2\rangle = |s_1\rangle, \quad \Omega^2 = \text{Id}.$$

4.2 Canonical representation of the fiber

The jump from a two-element set to a vector space is the point at which the paper is most vulnerable to the charge of importing QM-shaped tools. The following proposition makes the jump visible and earned.

Proposition 4.2 (Canonical faithful representation). *The minimal real linear representation that (i) faithfully carries the kernel action Ω (two distinct eigenvalues, ± 1) and (ii) faithfully carries the quarter-turn symmetry of the vibrational cycle is unique up to isomorphism, is two-dimensional, and admits a unique compatible complex structure.*

Proof. We proceed in four steps.

Step 1 (Dimension). The kernel $K = \ker(\pi) \cong \mathbb{Z}/2\mathbb{Z}$ acts freely and transitively on the two-element fiber F_b . Any faithful linear representation of $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{R}$ that distinguishes the two elements requires at least two linearly independent basis vectors (one per element).

A one-dimensional representation would identify Ω with ± 1 acting on $\mathbb{R}$, but then $|s_1\rangle$ and $|s_2\rangle = \Omega|s_1\rangle = -|s_1\rangle$ are scalar multiples of each other, losing the physical distinction between the two vibrational phases. Therefore $\dim_{\mathbb{R}} V_b \geq 2$.

A two-dimensional representation suffices: define $V_b = \text{span}_{\mathbb{R}}\{|s_1\rangle, |s_2\rangle\}$ with inner product $\langle s_i | s_j \rangle = \delta_{ij}$. The kernel acts as the permutation matrix $\Omega|s_1\rangle = |s_2\rangle$, $\Omega|s_2\rangle = |s_1\rangle$, which has eigenvalues $+1$ and -1 . No lower-dimensional faithful representation exists; therefore $\dim_{\mathbb{R}} V_b = 2$.

Step 2 (Quarter-turn). The vibrational cycle in scalar dimension 3 is a periodic motion (P2: two senses per scalar dimension; P4: vibrational occupation). Representing this cycle as planar rotation in $\mathbb{R}^2$, the two phases are separated by a half-turn (π radians). Define $J : V_b \rightarrow V_b$ as rotation by $\pi/2$. This is the unique orientation-preserving isometry of V_b satisfying:

$$J^2 = -\text{Id}, \quad \langle Jv | v \rangle = 0 \quad \text{for all } v \in V_b.$$

The uniqueness of J (up to sign, corresponding to orientation choice) follows from the fact that the only orthogonal transformations of $\mathbb{R}^2$ satisfying $J^2 = -\text{Id}$ are rotations by $\pm\pi/2$.

Step 3 (Complex structure). Define $i \cdot v := Jv$ for all $v \in V_b$. This satisfies $i^2 \cdot v = J^2 v = -v$, as required for the imaginary unit. Under this complex structure, $V_b \cong \mathbb{C}^1$ with the identification $|s_2\rangle = J|s_1\rangle = i|s_1\rangle$.

Step 4 (Uniqueness). The only linear operator on $\mathbb{R}^2$ that is (a) orthogonal, (b) satisfies $J^2 = -\text{Id}$, and (c) is compatible with the inner product ($\langle Jv | v \rangle = 0$) is $J = \pm R_{\pi/2}$ (rotation by $\pm\pi/2$). The two signs correspond to the two orientations of V_b , which are physically equivalent (choice of “which phase is first”). The complex structure is therefore canonical, not optional. $\square$

Corollary 4.3. *Complexification of the covering fiber via $J^2 = -\text{Id}$ is a structural consequence of the covering geometry and vibrational cycle. It is not an assumption imported from quantum mechanics.*

Status: T.

Remark 4.4 (Why complex numbers appear in the formalism). *The imaginary unit i is the representation-theoretic consequence of directional energy in the time region. Inverse speed (energy) in the time region is “directional” in the three-dimensional temporal frame (it has a phase from the vibrational cycle), and directional quantities in a 2D subspace are naturally represented by complex numbers.*

Remark 4.5 (Hurwitz consistency check). *The Hurwitz theorem [16] states that only $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$ admit a composition algebra. Since $\dim_{\mathbb{R}} V_b = 2$, $\mathbb{C}$ is the unique composition algebra of the correct dimension. This is a consistency check, not part of the derivation, but it confirms that no alternative to complex amplitudes exists for a 2D fiber.*

Remark 4.6 (Complex vs. quaternion wave functions). *Nehru [5] shows that the one-dimensional zone of the time region (electric sector, 1D rotation) requires complex wave functions $\phi = \{\phi_1, i\phi_2\}$, while the three-dimensional zone (magnetic sector, 2D rotation) requires quaternion wave functions $\psi = \{\psi_a, i\psi_c, j\psi_b, k\psi_d\}$ with Born-rule density $\psi^* \psi = \psi_a^2 + \psi_b^2 + \psi_c^2 + \psi_d^2$. This is consistent with the covering-fiber structure: the electric fiber $V_b \cong \mathbb{C}^1$ supports complex amplitudes, while the full magnetic structure (two 2D rotational dimensions) would require a quaternionic extension. The present paper works in the electric sector ($\mathbb{C}$); the quaternionic extension is identified as an open task (§16).*

5 Cross-Frame Projection

This is the heart of the paper. The derivation proceeds through four lemmas culminating in the cross-frame projection theorem. The dependency chain is:

Lem. 5.1 (π -uniqueness) $\rightarrow$ Lem. 5.3 (factorization) $\rightarrow$ Lem. 5.4 (linearity) $\rightarrow$
 Lem. 5.5 (Riesz) $\rightarrow$ Thm. 5.6 (projection).

5.1 Lemma 1: No electric–magnetic structural map exists other than π

Lemma 5.1 (π -channel uniqueness). *Within the imported compact fiber $F \cong \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$, the covering map $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ is the unique nontrivial group homomorphism from the electric sector to the magnetic sector.*

Proof. A group homomorphism $\phi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ is determined by the image of the generator: $\phi(1) \in \mathbb{Z}/4\mathbb{Z}$. For ϕ to be a homomorphism, $\phi(8 \cdot 1) = 8\phi(1) = 0$ in $\mathbb{Z}/4\mathbb{Z}$, which is satisfied for any choice. But ϕ must also satisfy $\phi(4) = 4\phi(1) = 0 \pmod{4}$, which is automatically satisfied. The nontrivial homomorphisms have $\phi(1) \in \{1, 2, 3\}$.

However, surjectivity requires $\phi(1)$ to generate $\mathbb{Z}/4\mathbb{Z}$, which means $\gcd(\phi(1), 4) = 1$, giving $\phi(1) \in \{1, 3\}$. These two maps are $\pi(k) = k \pmod{4}$ and $\pi'(k) = 3k \pmod{4}$. The map π' is the composition of π with the automorphism $x \mapsto 3x$ of $\mathbb{Z}/4\mathbb{Z}$, i.e., $\pi' = \alpha \circ \pi$ where α is an automorphism of the base. Since α is a relabeling of the magnetic sector (not a distinct structural map), both define the same covering structure up to automorphism of the base.

The remaining homomorphism $\phi(1) = 2$ has image $\{0, 2\} \subsetneq \mathbb{Z}/4\mathbb{Z}$; it is not surjective and does not connect the full magnetic sector. There is no other structural map.

Status: **T**. This is a straightforward computation in finite abelian group theory. $\square$

Remark 5.2. *Lemma 5.1 is stronger than what was claimed in draft 1, which only asserted that π -uniqueness was “physically compelling.” The proof is elementary. The uniqueness is not merely plausible; it is a theorem of the imported fiber structure.*

5.2 Lemma 2: Cross-frame events factor through π

Lemma 5.3 (Factorization). *Any admissible cross-frame readout—a process that translates information from the electric (temporal) sector to the magnetic (spatial) sector—must factor through π .*

Proof. The argument proceeds in two steps.

Step 1 (Admissible = group-respecting). By postulate P1, motion exists in discrete units. By the definition of displacement (Larson: deviations from unit speed), displacements combine by addition in $\mathbb{Z}/8\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$ respectively. A cross-frame readout $R : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ maps electric-sector displacements to magnetic-sector outcomes. If R failed to respect addition—that is, if $R(a + b) \neq R(a) + R(b)$ for some displacements a, b —then combining two displacement contributions a and b would produce a different spatial outcome than the sum of their individual outcomes. But the displacement algebra is additive: the physical combination of displacements a and b is $a + b$, not some other operation. A readout that violates additivity is therefore inconsistent with the displacement arithmetic of the postulates. Admissible readouts are group homomorphisms.

Step 2 (Uniqueness). By Lemma 5.1, π is the unique surjective group homomorphism $\mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ up to automorphism of the base. Any admissible readout therefore factors through π .

Testable consequence. If a non-additive cross-frame channel existed, spectroscopic terms of combined displacements would deviate from the additive displacement prediction. The quantum-defect law $\delta_l = [ab - l(l + 1)]/(2\sqrt{2})$, which is additive in the displacement product, works across 95 atom-observable pairs (Paper II). This empirically excludes non-additive channels.

Status: **T**. Step 1 follows from the additive nature of the displacement algebra (P1, definition of displacement). Step 2 is Lemma 5.1. The upgrade from **S** to **T** reflects the recognition that displacement additivity is not a modeling choice but a consequence of the postulates. $\square$

5.3 Lemma 3: After complexification, the induced readout is linear

Lemma 5.4 (Linearity). *After complexification (Proposition 4.2), the map induced by π on each fiber V_b is $\mathbb{C}$ -linear.*

Proof. The map $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ is a group homomorphism ($\pi(j+k) = \pi(j) + \pi(k)$). By the universal property of the group algebra, any group homomorphism $\pi : G \rightarrow H$ extends uniquely to a $\mathbb{C}$ -algebra homomorphism $\tilde{\pi} : \mathbb{C}[G] \rightarrow \mathbb{C}[H]$, defined by $\tilde{\pi}(\sum a_g g) = \sum a_g \pi(g)$. Restricted to the fiber $V_b \subset \mathbb{C}[\mathbb{Z}/8\mathbb{Z}]$, $\tilde{\pi}$ is a $\mathbb{C}$ -linear map $V_b \rightarrow \mathbb{C}$.

Status: T. This is a standard result in representation theory, applied to the already-derived complexification. $\square$

5.4 Lemma 4: Every nonzero linear readout is represented by a unique vector

Lemma 5.5 (Riesz representation on V_b). *Every nonzero $\mathbb{C}$ -linear functional $\ell : V_b \rightarrow \mathbb{C}$ is uniquely represented by a vector $|m\rangle \in V_b$ such that $\ell(|\psi\rangle) = \langle m|\psi\rangle$ for all $|\psi\rangle \in V_b$.*

Proof. Since $\dim_{\mathbb{C}} V_b = 1$, every nonzero linear functional is rank-1. By the Riesz representation theorem [14] on the inner-product space V_b (inner product derived in §6 below, or inherited from the translation-invariant metric of Paper I), there exists a unique $|m\rangle \in V_b$ with the stated property.

Status: T. Riesz's theorem is a mathematical fact applied to the inner-product space already derived. $\square$

5.5 Theorem: Cross-frame projection

Theorem 5.6 (Cross-frame projection). *Let V_b be the covering fiber at base point b , equipped with its inner product. Let $|\psi\rangle \in V_b$ be a normalized state ($\|\psi\| = 1$). Let a cross-frame event with measurement setting θ be represented by a unit vector $|m_\theta\rangle \in V_b$. Then the cross-frame map is the orthogonal projection*

$$P_\theta : V_b \rightarrow \mathbb{C} \cdot |m_\theta\rangle, \quad P_\theta|\psi\rangle = \langle m_\theta|\psi\rangle \cdot |m_\theta\rangle.$$

Proof. By Lemma 5.3, the readout factors through π . By Lemma 5.4, the induced map on V_b is $\mathbb{C}$ -linear. By Lemma 5.5, this linear functional is represented by a unique vector $|m_\theta\rangle$, so the outcome amplitude is $\langle m_\theta|\psi\rangle$, and the induced map $P_\theta|\psi\rangle = \langle m_\theta|\psi\rangle \cdot |m_\theta\rangle$ is the orthogonal projection onto $\mathbb{C} \cdot |m_\theta\rangle$.

Different measurement settings θ correspond to different choices of $|m_\theta\rangle$, which are rotations of a reference direction in V_b .

Status: T. All four lemmas are theorem-grade: π -uniqueness (Lem. 5.1), factorization from displacement additivity (Lem. 5.3), linearity from representation extension (Lem. 5.4), and Riesz representation (Lem. 5.5). $\square$

5.6 What the projection is and is not

The cross-frame projection is:

- *Non-invertible:* the kernel of π (the sheet distinction) is lost. This is the irreversibility.
- *Stochastic:* by the Born rule (derived below), the outcome probability is $|\langle m_\theta|\psi\rangle|^2$.
- *Not dynamical:* it is not a time evolution, not a Hamiltonian process, not a physical interaction. It is the structural relationship between two descriptions.

Unitarity governs the temporal frame (Schrödinger evolution); projection governs the frame crossing. These are distinct operations on distinct domains. No dynamical law changes at measurement. The apparent “collapse” is the dimension reduction from V_b (2D) to the outcome space (1D) through the covering map.

6 Born Rule and Probability Structure

The Born rule derivation proceeds through a chain of four lemmas culminating in Theorem 6.6:

Lem. 6.1 (linear expansion) $\rightarrow$ Lem. 6.2 ($p = 2$ uniqueness) $\rightarrow$ Lem. 6.4 (complex amplitudes forced) $\rightarrow$ Lem. 6.5 (phase gauge) $\rightarrow$ Thm. 6.6 ($P_i = |c_i|^2$).

6.1 Lemma 5: Linear expansion from bilinear integration

Lemma 6.1 (Linear expansion). *The coupling between branches in a composite system is computed by a bilinear integral. Therefore the state is a linear combination of branches: $\psi = \sum c_i B_i$.*

Proof. The coupling matrix element in the Förster system is:

$$\langle e_A g_B | V_{dd} | g_A e_B \rangle = \iint \psi_A^*(r_1) \psi_B^*(r_2) V(r_1, r_2) \psi'_A(r_1) \psi'_B(r_2) d^3 r_1 d^3 r_2.$$

This integral is bilinear in the wavefunctions—linear in each argument. Bilinearity is a property of integration, not an axiom of quantum mechanics.

The chain: bilinear integration $\rightarrow$ bilinear matrix elements $\rightarrow H$ is a linear operator $\rightarrow$ eigenstates are linear combinations $\rightarrow$ expansion $\psi = \sum c_i B_i$ is linear $\rightarrow$ coefficients c_i define a natural projection $\langle B_i | \psi \rangle = c_i$.

The linear expansion is also empirically confirmed: the Förster oscillation requires coherent (amplitude) mixing, not probabilistic (weight) mixing. A classical mixture would produce monotonic decay, not the observed oscillation.

Status: **T**. □

6.2 Lemma 6: Only $p = 2$ supports continuous dynamics

Lemma 6.2 (L^p uniqueness). *Among L^p -type norms, only $p = 2$ admits continuous norm-preserving linear dynamics.*

Proof. Given the linear expansion $\psi = \sum c_i B_i$ (Lemma 6.1) and the conservation law $\sum |c_i|^p = 1$, the set of linear maps preserving the L^p unit sphere forms a group. For different p :

p	Symmetry group	Continuous dynamics?
1	Permutations + sign changes	No (finite, no rotations)
2	O(n) rotations	Yes
> 2	Permutations only	No (no continuous subgroup)

Only $p = 2$ admits continuous rotations matching the observed Rabi oscillation in the Förster system. This is the mathematical content of the Pythagorean theorem: $\cos^2 \alpha + \sin^2 \alpha = 1$ for all α .

Numerical verification for a 2D rotation $c_1(t) = \cos(\omega t)$, $c_2(t) = \sin(\omega t)$:

p	$\min \sum c_i ^p$	$\max \sum c_i ^p$	Constant?
1.0	1.000	1.414	No
2.0	1.000	1.000	Yes
4.0	0.500	1.000	No

Status: T. The result that the L^2 unit sphere is the only one invariant under continuous linear isometries is a theorem of Lie group theory, applied to the branch space derived from Lemma 6.1. $\square$

Corollary 6.3 (Inner product). *With $p = 2$ established, the L^2 norm $\|\psi\|^2 = \sum c_i^2 = 1$ is conserved. The inner product follows by polarization:*

$$\langle \psi | \phi \rangle = \frac{1}{4} [\|\psi + \phi\|^2 - \|\psi - \phi\|^2] = \sum c_i(\psi) c_i(\phi).$$

Status: T.

6.3 Lemma 7: Oscillatory dynamics forces complex amplitudes

Lemma 6.4 (Complexification forced by dynamics). *Real-valued branch weights cannot account for oscillatory dynamics. Complex amplitudes are required for norm-preserving time evolution.*

Proof. The Förster system starts at $t = 0$ with real expansion coefficients: $|\psi(0)\rangle = \cos \theta |+\rangle + \sin \theta |-\rangle$. Time evolution introduces phases:

$$|\psi(t)\rangle = \cos \theta e^{-iE_+t} |+\rangle + \sin \theta e^{-iE_-t} |-\rangle.$$

The phases are physical—they generate the observed oscillation:

$$P(dd, t) = \cos^4 \theta + \sin^4 \theta + 2 \cos^2 \theta \sin^2 \theta \cos(\Delta E t / \hbar).$$

If we tried to keep only real parts of the coefficients, normalization $\sum |c_i|^2 = 1$ would not be preserved at all times. Complex amplitudes are therefore required.

This is CG2 (complexification): the promotion of real selector weights $W_i \geq 0$ to complex amplitudes $c_i \in \mathbb{C}$.

Status: T. The argument is a direct computation using the Förster dynamics. $\square$

6.4 Lemma 8: Phase gauge invariance removes phase dependence

Lemma 6.5 (Gauge invariance). *The weight function $f(c)$ depends only on $|c|$, not on the phase of c .*

Proof. Write $c = re^{i\phi}$ in polar form. The weight function is $f(c) = g(r, \phi)$. Multiplying all amplitudes by an overall phase $e^{i\alpha}$ (corresponding to shifting the energy zero, an arbitrary convention) must leave $\sum f(c_i)$ unchanged. Taking the derivative with respect to α :

$$\sum_i \frac{\partial g}{\partial \phi}(r_i, \phi_i + \alpha) = 0 \quad \text{for all } \alpha.$$

Since this must hold for all configurations $\{r_i, \phi_i\}$, each term vanishes independently: $\partial g / \partial \phi = 0$ everywhere. Therefore g depends only on $r = |c|$.

Status: T. $\square$

6.5 Theorem: The Born rule

Theorem 6.6 (Born rule). *The unique admissible branch weight is $P_i = |c_i|^2$.*

Proof. Step 1. From Lemma 6.2, the pre-complexification weight rule is $W_i = c_i^2$ for real positive c_i (the squared projection, Pythagorean decomposition).

Step 2. From Lemma 6.4, the dynamics requires promotion to complex amplitudes.

Step 3. From Lemma 6.5, $f(c)$ depends only on $|c|$.

Step 4. Since f depends only on $|c|$, and for real positive c we have $f(c) = c^2$ from Step 1, therefore $f(c) = |c|^2$ for all complex c .

Step 5. Parseval normalization: $\sum |c_i|^2$ is preserved under unitary evolution (the definition of unitarity), confirming that the Born rule satisfies normalization at all times.

*Status: **S/T**.* The conditional derivation (given the three premises) is **T**. Overall status is **S** because Premise 1 (the pre-CG2 squared-projection weight rule) depends on the inner product (Corollary 6.3), which depends on the linear branch expansion (Lemma 6.1). All dependencies are themselves **T**; the **S** reflects only the multi-step conditional structure. $\square$

6.6 Second-power uniqueness: three independent routes

The exponent $p = 2$ is forced by three independent routes, all converging on the same structural fact: (i) conservation + continuity selects the unique L^2 norm admitting continuous rotations (Lemma 6.2); (ii) the inter-regional relation $r = \rho^2$ requires $p = 2$ for dimensional consistency of the density $|\psi(\sqrt{r})|^p \times |d\rho/dr|$; (iii) numerical Förster evolution preserves normalization only at $p = 2$:

t (ns)	$p = 2$ (Born)	$p = 1$	$p = 4$
0	1.000	1.000	1.000
50	1.000	1.404	0.528
125	1.000	1.408	0.517
250	1.000	1.166	0.935

The L^2 norm is the unique inner-product norm; inner-product norms are the unique norms compatible with continuous rotational dynamics. The convergence of three routes is not coincidental—it is three views of one fact.

6.7 Gleason-type consistency

The covering fiber $V_b \cong \mathbb{C}^1$ has complex dimension 1, below the Gleason threshold (dimension ≥ 3) [10]. The derivation above bypasses this via Parseval and second-power uniqueness. However, the full electric representation space $V_{\text{elec}} = \bigoplus_{b \in \mathbb{Z}/4\mathbb{Z}} V_b \cong \mathbb{C}^4$ has dimension $4 \geq 3$, so Gleason’s theorem applies there and confirms $P = |\langle v|\Psi\rangle|^2$ as the unique σ -additive probability measure.

The two results are complementary: the second-power relation fixes $p = 2$ on dimension 2 (where Gleason is silent); Gleason fixes $p = 2$ on dimension ≥ 3 (where the full fiber lives). Neither is redundant.

6.8 Completeness bridge to continuous observables

Corollary 6.7 (Extension to continuous observables). *The discrete Born rule $P(n) = |c_n|^2$ extends to continuous position measurements via Parseval’s theorem: $\int |\psi(\rho)|^2 d\rho = \sum_n |c_n|^2 = 1$, so $|\psi(\rho)|^2 d\rho$ is a probability density. The KS mapping $r = \rho^2$ transforms this to the spatial Born rule. The momentum Born rule follows from the Plancherel theorem.*

*Status: **T**.* The three ingredients beyond Theorem 6.6—completeness, Parseval, Plancherel—are mathematical theorems, not physical postulates.

7 Effective Hilbert Structure

7.1 The three coarse-grainings

The tensor-product Hilbert space is not assumed—it is derived from the selector structure through three physical coarse-grainings, each with a well-defined domain of validity:

CG1: Discretization (continuous branches $\rightarrow$ discrete set). Each atom’s continuous branch geometry is truncated to the relevant energy levels. *Valid when* the relevant branches are well-separated in energy compared to the coupling V . *Gained:* finite-dimensional algebra. *Lost:* sub-branch structure, continuous spectral detail.

CG2: Complexification (real weights $\rightarrow$ complex amplitudes). The selector weights $W_i \geq 0$ (probability simplex) are promoted to complex amplitudes $c_i \in \mathbb{C}$ (unit sphere). *Valid when* decoherence is slow compared to the coupling timescale $\hbar/V$. *Gained:* phases, interference, unitary time evolution. *Lost:* positive-definiteness of weights.

CG3: Tensor product (Cartesian product $\rightarrow \otimes$ product). The composite branch set $\mathcal{B}_A \times \mathcal{B}_B$ is promoted to $\mathcal{H}_A \otimes \mathcal{H}_B$. *Valid when* electron overlap between A and B is negligible. *Gained:* entanglement, Bell states. *Lost:* sub-system mixing (exchange effects).

The Hilbert-space description is the special case of the selector framework that applies when all three coarse-grainings are valid simultaneously.

7.2 The four-level structure

Level	Content	Mathematical structure
0: Carrier geometry	Angular bundle + radial sector	Continuous fiber bundle
1: Selector law	Gate $\times$ weight $\times$ scale	Empirical law on geometry
2: Composite selector	Signed superpositions of products	Discretized, real-valued mixing
3: Hilbert space	$\mathcal{H}_A \otimes \mathcal{H}_B$, complex amplitudes	After CG1 + CG2 + CG3

The selector framework provides Levels 0–2. The three coarse-grainings produce Level 3 from Level 2.

7.3 Superposition

A state $|\psi\rangle = c_1|s_1\rangle + c_2|s_2\rangle$ describes a scalar-motion entity that occupies both sheets of the covering fiber simultaneously. The two vibrational phases coexist; the entity is in both phases at once, as a vibrating string occupies both its upward and downward excursions simultaneously. Superposition is the natural state of a vibrational entity; “collapse” to a single outcome is the projection from the fiber to its base. *Status: I.*

7.4 Inner product

The inner product on V_b is derived from three physical inputs via Lemma 6.2 and Corollary 6.3. *Status: T.*

7.5 Tensor products

For two spatially separated systems A and B , the joint state space is $V_A \otimes V_B$. This is CG3, valid when electron overlap is negligible. The composite selector law on $\mathcal{B}_A \times \mathcal{B}_B$ induces the algebraic tensor product once overlap vanishes: spatial independence means independent branch sets, and the tensor product is the unique bilinear extension of the Cartesian product of branches to a vector space. When overlap is significant, CG3 breaks down and the tensor-product representation fails; the selector framework (Levels 0–2) still applies, but Level 3 does not. *Status: T under the overlap condition already required for Level-3 validity.*

7.6 Unitary evolution

Time evolution in the temporal frame is governed by the Schrödinger equation, producing unitary evolution $|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$. Unitarity preserves the L^2 norm: $\|\psi(t)\| = \|\psi(0)\|$. This is the temporal-frame analog of energy conservation: unitary evolution preserves the total

selector weight. *Status: I* (identification of Schrödinger evolution as time-region mechanics, following Nehru).

7.7 What Hilbert space preserves and discards

Preserves: interference, the Born-rule probability structure, Bell correlations, unitary time evolution, the superposition principle.

Discards:

1. The displacement numbers (a, b, c) that determine the time-region potential.
2. The covering-map structure (source/screen decomposition, covering degree, kernel action).
3. The quantum-defect law $\delta_l = [ab - l(l+1)]/(2\sqrt{2})$.
4. The selector hierarchy (blocking, additive, relational, mixed angular-radial).

Standard QM compensates for these losses by introducing empirical parameters (configuration-interaction coefficients, screening constants) that the RST framework derives from the displacement algebra.

8 Wave–Particle, Measurement, and Preferred Basis

8.1 Dissolution of wave–particle duality

Wave–particle duality dissolves: the wave and the particle are not two entities but two descriptions of one scalar-motion entity, viewed from reciprocal reference frames (§3). *Status: T*.

8.2 Why collapse is projection, not dynamical rupture

Unitarity governs the temporal frame; cross-frame projection governs the frame crossing. These are distinct operations on distinct domains. No dynamical law changes at measurement. The cross-frame projection is non-invertible (the kernel is lost), stochastic (Born rule), and not dynamical (structural relationship, not time evolution). *Status: S, conditional on Theorem 5.6*.

8.3 Preferred basis from detector coupling

Proposition 8.1 (Preferred basis). *The detector-system coupling has the form*

$$V_{SD} = \sum_i g_i |i\rangle\langle i| \otimes T_i,$$

where $|i\rangle\langle i|$ projects onto system branch i , g_i is the coupling strength, and T_i is the detector transition triggered by branch i . This coupling is diagonal in the system branch basis. After coupling and decoherence ($\langle D_i | D_j \rangle \rightarrow 0$ for $i \neq j$), the reduced system density matrix is:

$$\rho_S = \sum |c_i|^2 |i\rangle\langle i|.$$

In any other basis, ρ_S has off-diagonal elements. The preferred basis is determined by the physics of the coupling, not by the observer's choice.

Proof. After coupling, the joint state is $|\Psi\rangle = \sum c_i |i\rangle \otimes |D_i\rangle$ where $|D_i\rangle = T_i |D_0\rangle$. Tracing over the detector:

$$\rho_S = \text{tr}_D(|\Psi\rangle\langle\Psi|) = \sum_{i,j} c_i c_j^* \langle D_j | D_i \rangle |i\rangle\langle j|.$$

After decoherence ($\langle D_j | D_i \rangle \rightarrow \delta_{ij}$), this reduces to $\rho_S = \sum |c_i|^2 |i\rangle\langle i|$.

Status: T.

□

This coupling law follows the same gate/weight/scale structure as all previous selector laws:

Stage	Gate	Weight	Scale
3A (blocking)	$\Theta(\Delta\sigma = +1)$	$ \Delta C $	$\sigma = 1/d$
3E (Förster)	$\kappa(\Omega)$	$\mu_1\mu_2/\delta$	R^{-3}
Measurement	$ i\rangle\langle i $	g_i	T_i

Three ingredients produce stable branch basins: (a) diagonal coupling (no branch mixing), (b) macroscopic amplification ($\langle D_i | D_j \rangle \approx 0$), (c) thermodynamic irreversibility. None requires a measurement axiom.

8.4 The three levels of the measurement question

Level	Question	Status
A	Why definite values?	T : decoherence (CG2 breakdown) produces diagonal ρ .
B	Why $ c ^2$ probabilities?	T : Born rule from Lem. 6.2–6.5, Thm. 6.6.
C	Why THIS outcome?	S/I : structurally dissolved (see below).

8.5 Structural dissolution of single-outcome selection

The measurement problem in standard QM arises from four premises:

- (a) The state space (Hilbert space) is the entire ontology.
- (b) The state is in a superposition before measurement.
- (c) Measurement produces a definite single outcome.
- (d) No mechanism within (a) explains the transition (b)→(c).

In the RST framework, premise (a) fails. The ontologically primitive state space is the *fiber*, not Hilbert space. Hilbert space is an effective projection of the fiber after CG1–CG3 (§7.1), and the projection discards information:

- (a') The state space is the fiber V_b , which carries more structure than the Hilbert-space projection retains.
- (b') The fiber state is definite at all times. The superposition is the spatial-frame *description* of the fiber state, not the fiber state itself. Superposition is a property of the projection, not of the ontology.
- (c') The cross-frame projection through π (Theorem 5.6) maps the fiber state to a spatial outcome.
- (d') The projection is a well-defined structural operation. Its stochasticity is a consequence of discarding the kernel component of the fiber state, which is structurally inaccessible from the spatial frame.

The dissolution is precise: the covering map $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ has kernel $\ker(\pi) = \{0, 4\}$. A state in the fiber has a *base component* (what π preserves, mapping to the spatial outcome) and a *kernel component* (what π destroys). The spatial observer sees only the base component. The apparent randomness of the outcome is not ontological indeterminacy but *projective information loss*: the kernel information that would determine the outcome is carried by the fiber but is structurally inaccessible from the spatial reference frame.

Proposition 8.2 (Projective origin of apparent randomness). *Let $|\psi\rangle \in V_b$ be a definite fiber state, and let P_θ be the cross-frame projection (Theorem 5.6). The outcome probability $p(\theta) = |\langle m_\theta | \psi \rangle|^2$ is fully determined by the fiber state $|\psi\rangle$ and the measurement setting θ . The spatial observer, who has access only to the projected (base) component of the fiber, sees a probability distribution rather than a definite outcome because the kernel component that would resolve the ambiguity is not accessible through any spatial measurement.*

Proof. The cross-frame projection through π discards the kernel component $\ker(\pi) = \{0, 4\}$ while preserving the base component (Theorem 5.6). The fiber state $|\psi\rangle$ is definite; the spatial observer sees only the base projection, yielding Born probability $|\langle m_\theta | \psi \rangle|^2$ that is kernel-insensitive ($|\langle m_\theta | - \psi \rangle|^2 = |\langle m_\theta | \psi \rangle|^2$) because $\Omega = -\text{Id}$ on V_b (Proposition 4.2). The Born rule $p(\theta) = |\langle m_\theta | \psi \rangle|^2$ is the squared modulus precisely because it must be insensitive to the kernel phase.

Status: **S/I**. The structural argument (projection discards kernel, Born rule is kernel-insensitive) is theorem-grade. The interpretive identification (the fiber state is ontologically definite, the randomness is projective) is **I**. $\square$

Remark 8.3 (Comparison with other interpretations). *The RST dissolution differs from all standard interpretations:*

<i>Interpretation</i>	<i>Position on single-outcome selection</i>
<i>Copenhagen</i>	<i>Measurement axiom (postulated, not derived)</i>
<i>Many-worlds</i>	<i>Both outcomes occur; single outcomes are apparent</i>
<i>Bohmian</i>	<i>Spatial hidden variable (particle position) determines outcome</i>
<i>QBism</i>	<i>Probabilities are subjective; no fact of the matter</i>
<i>RST (this paper)</i>	<i>Fiber state is definite; randomness is projective information loss through π; kernel component is structurally inaccessible from spatial frame</i>

The RST position is closest to Bohmian mechanics in that the underlying state is definite, but differs in that the hidden information is temporal (fiber/kernel component) rather than spatial (particle position). This distinction matters for Bell compatibility: the fiber variables are non-spatial and do not enter Bell’s locality assumption.

Remark 8.4 (Bell compatibility). *This dissolution does not constitute a local hidden-variable model. Bell’s theorem [12] rules out models where a shared classical variable λ produces factorizable correlations $P(a, b) = \int P(a|\theta_A, \lambda) P(b|\theta_B, \lambda) d\lambda$. In the present framework, the shared fiber state $|\Psi\rangle$ (the singlet, Lemma 9.1) is entangled—its correlations are non-factorizable by construction. The fiber state is not a classical hidden variable; it is the ontologically primitive quantum state viewed from the fiber rather than from Hilbert space. The framework is spatially local (each projection acts at one detector) and temporally non-separable (the entangled pair shares a common fiber position from pair creation). This is not spatial nonlocality; it is temporal adjacency in the time region, which has no spatial separation.*

Remark 8.5 (What remains). *The dissolution explains why single outcomes occur (the fiber state is definite) and why they appear random from the spatial frame (kernel information loss through π). It does not predict which outcome occurs in a specific run. Predicting individual outcomes would require solving the time-region dynamics for the kernel component, which is a question about the completeness of the time-region mechanics—not about the measurement process itself. The residual question is whether the time-region dynamics is deterministic or fundamentally stochastic at the fiber level. The postulates do not resolve this; it is identified as a question for the research program, not as a defect of the measurement analysis.*

8.6 Decoherence as CG2 breakdown

Decoherence is the process by which the environment becomes correlated with the system's branch choice, destroying the phase relationship. The quantitative criterion is $\eta = T_2/T_{\text{osc}}$, where T_2 is the decoherence time and T_{osc} the oscillation period:

η	Regime	Description
$\eta \gg 1$	CG2 valid (quantum)	Coherent oscillation, Born rule, entanglement
$\eta \sim 1$	CG2 marginal	Damped oscillation, partial coherence
$\eta \ll 1$	CG2 broken (classical)	No oscillation, selector weights only

For the Ravets et al. Förster experiment: $T_{\text{osc}} \approx 170$ ns, $T_2 \approx 3$ μ s, giving $\eta \approx 17$ —CG2 well satisfied.

The density matrix transitions:

$$\rho = \begin{pmatrix} |c_1|^2 & c_1 c_2^* \\ c_2 c_1^* & |c_2|^2 \end{pmatrix} \xrightarrow{t \gg T_2} \rho = \begin{pmatrix} W_1 & 0 \\ 0 & W_2 \end{pmatrix}.$$

Decoherence is the return from Level 3 to Level 1. *Status: **T** for the mechanism; **I** for the identification with CG2 breakdown.*

Remark 8.6 (FRET as the classical limit). *FRET is the $\eta \ll 1$ limit: Rabi oscillations ($P = \sin^2(Vt)$, $V \propto R^{-3}$) become monotonic transfer ($P = V^2 t / \Gamma$, $k \propto R^{-6}$). The R^{-6} FRET scaling is the classical shadow of the R^{-3} coupling, via Fermi's golden rule.*

8.7 What remains effective rather than derived

The measurement-conditioned weight update ($W_j \rightarrow \delta_{ij}$ on observing outcome i) is Bayesian conditioning—interpretation-neutral, the same in classical and quantum probability. The boundary between derived (everything operational) and underdetermined (single-outcome selection) is the boundary between operational physics and interpretive metaphysics.

9 Bell Correlations and Covering Coherence

9.1 Why Bell belongs here

Bell's theorem [12] and the CHSH inequality [13] are the sharpest tests of any framework that aims to derive quantum structure. This section derives the correlation functions through two lemmas and a theorem.

9.2 Antisymmetric state selection

Lemma 9.1 (Antisymmetric state from datum neutrality). *For a photon pair created from a single vibration, P5 (datum neutrality) requires the pair state to be annihilated by the total quarter-turn generator $J_{\text{total}} = J_A \otimes \text{Id}_B + \text{Id}_A \otimes J_B$. The unique such state (up to complex scalar) is the antisymmetric singlet:*

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|s_1\rangle_A |s_2\rangle_B - |s_2\rangle_A |s_1\rangle_B).$$

Proof. Step 1 (P5 constraint). Postulate P5 states that the shared datum is the neutral condition relative to which all admissible structural invariants vanish. A photon pair from a single vibration can be annihilated (restored to datum) only if every fiber invariant of the pair state

vanishes. The quarter-turn generator J (the infinitesimal form of the vibrational cycle, Proposition 4.2) is the structural invariant of the covering fiber. For the pair, the relevant generator is:

$$J_{\text{total}} = J_A \otimes \text{Id}_B + \text{Id}_A \otimes J_B,$$

which measures the total vibrational content across both subsystems. Datum neutrality requires $J_{\text{total}}|\Psi\rangle = 0$.

Step 2 (Uniqueness). The operator J_{total} acts on $V_A \otimes V_B \cong \mathbb{R}^4$. Using $J|s_1\rangle = |s_2\rangle$ and $J|s_2\rangle = -|s_1\rangle$, we compute J_{total} on each basis element:

$$\begin{aligned} J_{\text{total}}|s_1 s_1\rangle &= |s_2 s_1\rangle + |s_1 s_2\rangle, \\ J_{\text{total}}|s_1 s_2\rangle &= |s_2 s_2\rangle - |s_1 s_1\rangle, \\ J_{\text{total}}|s_2 s_1\rangle &= -|s_1 s_1\rangle + |s_2 s_2\rangle, \\ J_{\text{total}}|s_2 s_2\rangle &= -|s_1 s_2\rangle - |s_2 s_1\rangle. \end{aligned}$$

Direct computation:

$$J_{\text{total}}(|s_1 s_2\rangle - |s_2 s_1\rangle) = (|s_2 s_2\rangle - |s_1 s_1\rangle) - (-|s_1 s_1\rangle + |s_2 s_2\rangle) = 0.$$

Over $\mathbb{R}$, the null space of J_{total} is two-dimensional, spanned by $|s_1 s_2\rangle - |s_2 s_1\rangle$ and $J(|s_1 s_2\rangle - |s_2 s_1\rangle)$. Under the complex structure (Proposition 4.2), these two real vectors are the real and imaginary parts of a single complex vector. The null space over $\mathbb{C}$ is therefore one-dimensional: the antisymmetric state $|\Psi\rangle$, unique up to overall phase.

The three symmetric states $|s_1 s_1\rangle$, $|s_2 s_2\rangle$, and $(|s_1 s_2\rangle + |s_2 s_1\rangle)/\sqrt{2}$ are verified to be *not* annihilated by J_{total} .

Testable predictions. The singlet state makes two experimentally distinguishable predictions:

1. *Correlation form:* $E(\theta_A, \theta_B) = -\cos 2(\theta_A - \theta_B)$.
2. *Rotational invariance:* E depends only on the angle difference $\theta_A - \theta_B$, not on the individual angles.

The symmetric triplet state $(|s_1 s_2\rangle + |s_2 s_1\rangle)/\sqrt{2}$ violates rotational invariance: its correlation function depends on the absolute orientation of the measurement axes. Every Bell experiment confirms both the $-\cos 2\theta$ form and the rotational invariance, selecting the singlet over all triplet alternatives.

Status: T. Step 1 follows from P5 applied to the quarter-turn generator. Step 2 is a direct computation on $\mathbb{R}^4$. The testable predictions are confirmed experimentally. $\square$

9.3 Bell correlation theorem

Theorem 9.2 (Bell correlations). *The photon-polarization correlation function is $E(\theta) = -\cos 2\theta$, and the CHSH value at optimal angles is $|S| = 2\sqrt{2}$, saturating the Tsirelson bound [11].*

Proof. For photon polarization with measurement basis at angle θ :

$$\begin{aligned} |m_+(\theta)\rangle &= \cos \theta |s_1\rangle + \sin \theta |s_2\rangle, \\ |m_-(\theta)\rangle &= -\sin \theta |s_1\rangle + \cos \theta |s_2\rangle. \end{aligned}$$

Computing the four joint probabilities for settings $\theta_A = 0$, $\theta_B = \theta$:

$$\begin{aligned} P(+, +) &= \frac{1}{2} \sin^2 \theta, & P(+, -) &= \frac{1}{2} \cos^2 \theta, \\ P(-, +) &= \frac{1}{2} \cos^2 \theta, & P(-, -) &= \frac{1}{2} \sin^2 \theta. \end{aligned}$$

The correlation: $E(\theta) = \sin^2 \theta - \cos^2 \theta = -\cos 2\theta$.

At CHSH-optimal angles $(\theta_1, \theta_2, \theta_3, \theta_4) = (0, \pi/4, \pi/8, 3\pi/8)$: $|S| = 2\sqrt{2}$.

Status: T. The singlet state is derived from P5 (Lemma 9.1). The computation is exact. $\square$

9.4 Spin- $\frac{1}{2}$: $E(\theta) = -\cos \theta$

The magnetic sector supports two-dimensional rotation. As shown by Nehru [5], the distinction between spin-1 and spin- $\frac{1}{2}$ follows from the dimensionality of the rotation:

Lemma 9.3 (Half-angle measurement basis from 2D rotation). *One-dimensional rotation (the electric sector) cycles at 2π radians; the quantum of angular momentum is $\hbar$ (spin-1). Two-dimensional rotation (the magnetic sector) cycles at 4π steradians; the quantum of angular momentum is $\frac{1}{2}\hbar$ (spin- $\frac{1}{2}$). The measurement basis for spin- $\frac{1}{2}$ particles is therefore at half-angle: $\cos(\theta/2)$, $\sin(\theta/2)$.*

Proof. The momentum of a rotation is energy divided by rotational speed. For one-dimensional rotation, the speed unit is 2π radians per unit time, giving the angular momentum quantum $\hbar/(2\pi) = \hbar$ (spin-1, one full cycle at 2π). For two-dimensional rotation, the speed unit is 4π steradians per unit time (the full solid angle of a sphere), giving the angular momentum quantum $\hbar/(4\pi) = \frac{1}{2}\hbar$ (spin- $\frac{1}{2}$, one full cycle at 4π).

The 4π cycle means a 2π rotation returns to the starting state with a sign change, and a second 2π rotation is needed to restore the original sign. A measurement that projects the 2D rotation onto a spatial axis at angle θ therefore operates at half the angular rate of the 1D case: the measurement basis vectors are $\cos(\theta/2)$ and $\sin(\theta/2)$ rather than $\cos \theta$ and $\sin \theta$.

Status: T. The argument follows from the distinction between 1D and 2D rotational space (P2: two senses per scalar dimension, applied to 1 vs. 2 magnetic dimensions). The 4π -steradian interpretation is Nehru's; the half-angle consequence is a direct geometric result. $\square$

The singlet state (Lemma 9.1) with the half-angle basis gives:

$$P(\pm, \pm) = \frac{1}{2} \sin^2(\theta/2), \quad P(\pm, \mp) = \frac{1}{2} \cos^2(\theta/2), \quad E(\theta) = -\cos \theta, \quad |S| = 2\sqrt{2}.$$

Status: T. The singlet is derived from P5 (Lemma 9.1), the half-angle basis from the 2D rotation cycle (Lemma 9.3), and the computation is exact.

9.5 Tsirelson bound from covering degree

Proposition 9.4 (Tsirelson bound). *The covering-coherent state doubles the classical-mixture correlation amplitude. The amplification factor is $d = 2$ (covering degree).*

Classical mixture: If the sheet assignment is a classical hidden variable $\lambda \in [0, \pi)$:

$$E_{\text{mix}}(\theta) = -\frac{1}{2} \cos(2\theta), \quad |S_{\text{mix}}| = \sqrt{2} \leq 2.$$

Covering-coherent state: The coherent amplitude sum over both sheets produces $E(\theta) = -\cos(2\theta)$, doubling the amplitude. At CHSH-optimal angles: $|S| = 2\sqrt{2}$.

The value $d = 2$ follows from P4 (one vibrational dimension with two phases), the minimal nontrivial covering. *Status: S, conditional on Theorem 9.2.*

9.6 What is and is not being claimed

- This is *not* a local hidden-variable model. The covering fiber is not a classical hidden variable.
- This is *not* a denial of standard quantum predictions. The correlation functions are identical to standard QM.
- This *is* a structural derivation of the admissibility bound from the postulates. The singlet state is uniquely selected by P5 (datum neutrality) via $J_{\text{total}} = 0$ (Lemma 9.1). The Tsirelson value $2\sqrt{2}$ arises from the covering degree $d = 2$, which is derived from the postulates.

- The derivation makes two testable predictions beyond the standard Bell test: (a) the singlet form $E = -\cos 2(\theta_A - \theta_B)$, and (b) rotational invariance (dependence on angle *difference* only). The symmetric triplet alternative violates (b). Both predictions are confirmed experimentally.

10 Quantitative Anchor: The Förster Test

The Förster experiment is useful as a regime in which all three coarse-grainings are simultaneously valid, and therefore the effective Hilbert representation should apply quantitatively. It does not prove the abstract claims of the paper; it validates that the conditions for those claims are met in a specific physical system.

10.1 The experimental system

Ravets et al. [17] observed coherent dipole-dipole oscillations between two ^{87}Rb atoms at the Stark-tuned Förster resonance:

$$|dd\rangle = |59D_{3/2}, 59D_{3/2}\rangle \rightarrow |pf\rangle = |61P_{1/2}, 57F_{5/2}\rangle.$$

At resonance, $V = C_3/R^3$ with measured $C_3 = 2.1 \pm 0.1 \text{ GHz} \cdot \mu\text{m}^3$.

10.2 The composite selector law

The composite branches are:

$$\begin{aligned} |+\rangle &= \cos \theta |e_A g_B\rangle + \sin \theta |g_A e_B\rangle, \\ |-\rangle &= -\sin \theta |e_A g_B\rangle + \cos \theta |g_A e_B\rangle, \end{aligned}$$

where $\tan(2\theta) = V/\delta$ and δ is the energy detuning. The mixing angle decomposes as:

$$\tan(2\theta) = \underbrace{\kappa(\Omega)}_{\text{gate}} \times \underbrace{\mu_A \mu_B / \delta}_{\text{weight}} \times \underbrace{R^{-3}}_{\text{scale}}.$$

Entanglement is the maximally non-factorizable limit ($\theta = \pi/4$, exact resonance $\delta = 0$).

10.3 Regime validation

CG	Condition	Status
CG1	Pair states well-separated	✓ (isolated Stark channel)
CG2	Coherent evolution	✓ ($\eta \approx 17$)
CG3	No electron overlap	✓ ($R = 9\text{--}15 \mu\text{m} \gg 0.2 \mu\text{m}$)

Status: E. This is regime validation, not proof. The experiment confirms that the three coarse-grainings are simultaneously satisfied and the Hilbert representation applies quantitatively.

10.4 The R^{-3} scaling

R (μm)	f_{obs} (MHz)	f_{pred} (MHz)	Ratio
9	~ 3.0	2.9	0.96
11	~ 1.5	1.6	1.05
13	~ 1.0	1.0	0.96
15	~ 0.6	0.6	1.04

Predictions agree with measurements to within $\sim 5\%$ distance calibration uncertainty.

11 Identical Particles and the Pauli Principle

11.1 Reference-point structure and exchange symmetry

Two particles occupy the same reference point on the compact fiber when they share the identical displacement triple (a, b, c) . For such particles, spatial exchange is algebraically identical to the kernel action Ω on each fiber (Lemma 5.1): both operations correspond to the sheet-switching involution that the covering map π already defines. For half-integer-spin particles (2D rotation, 4π cycle per Lemma 9.3), $\Omega = -\text{Id}$ in the rotational representation (Proposition 4.2), so the admissible joint state must be antisymmetric under exchange. For integer-spin particles (1D rotation, 2π cycle), the exchange introduces no sign, and the admissible state is symmetric.

Status: T. The algebraic identification follows directly from the kernel action (Lemma 5.1) and the rotational representation (Proposition 4.2). No additional identification step is needed.

11.2 Pauli exclusion as emergent restriction

Proposition 11.1 (Pauli exclusion). *If two identical fermions occupied the same single-particle state $|\phi\rangle$, the joint state $|\phi\rangle \otimes |\phi\rangle$ would be symmetric under exchange, violating the antisymmetry constraint. Therefore identical fermions must occupy distinct states.*

Example. Consider two electrons (displacement 0-0-(1), spin- $\frac{1}{2}$) in the same atom. Both share the same fiber position and reference point. The antisymmetry requirement forces them into distinct single-particle states—different (n, l, m_l, m_s) quantum numbers. This is the Pauli exclusion principle, which in the RST is not an independent postulate but a consequence of the covering-sheet exchange structure applied to 2D rotational (half-integer-spin) entities.

The mathematical consequence (Pauli exclusion given antisymmetry) is **T**; the physical identification of spatial exchange with sheet-switching is **S**.

12 Relation to Standard Quantum Mechanics

12.1 Where the framework reproduces standard QM

Within the domain of validity of the three coarse-grainings, the present framework and standard QM make identical predictions: energy eigenvalues, Born-rule probabilities, Bell correlations, decoherence rates, superposition, interference, and entanglement.

12.2 Where it is interpretive

QM primitive	RST origin	Status
Complex amplitudes	Vibrational phase space (P2 + P4)	T
Superposition	Both sheets of covering fiber	I
Inner product	Bilinearity + conservation + continuity	T
Born rule $P = \psi ^2$	L^2 uniqueness + gauge invariance	S/T
Measurement (“collapse”)	Cross-frame projection through π	T
Tensor products	Selector algebra + vanishing overlap	T
Unitary evolution	Schrödinger equation in time region	I
Bell violation	Covering coherence, degree 2	T
Pauli exclusion	Kernel action + rotational representation	T

In contrast to purely information-theoretic derivations of quantum formalism, the present framework begins from physical postulates about discrete scalar motion and derives both effective quantum structure and effective gravitational structure from the same compact fiber that produces the periodic table in Paper II.

12.3 What standard QM treats as primitive that this paper treats as effective

- The Coulomb potential $V = -e^2/r$ (KS image of the harmonic progression potential; derived in Paper II).
- Hilbert space as fundamental state space (effective representation after CG1–CG3).
- The Born rule as an axiom (derived from Lemmas 6.2–6.5).
- Wave-function collapse as a dynamical process (covering-to-base projection through π).
- The Pauli exclusion principle as an independent postulate (from kernel action on the covering fiber).
- Electron-electron interaction integrals as primitive inputs (effective parameters compensating for Hilbert space’s information loss).

12.4 Scope of this paper within the RST program

This paper derives effective quantum structure and effective gravitational structure from the compact fiber. It does not address the detailed sub-atomic particle classification, charge structure, or the strong nuclear interaction already developed in the broader RST program [3]. The following are therefore outside the scope of *this paper*, not outside the scope of the RST:

- The detailed sub-atomic particle classification (Larson, points 63–79 of [3]).
- Charge structure and electrostatic/electromagnetic phenomena (Larson, points 98–142).
- The strong nuclear interaction and nuclear structure.
- Gravitational field equations (the effective metric is derived here, but the field equation it satisfies is not; see §16).

The connection between the compact-fiber formalism of Papers I–III and the broader RST particle program is a natural target for future work. What the present trilogy demonstrates is that the foundational layer—the compact fiber, the covering map, and the displacement algebra—supports both quantitative atomic predictions (Paper II) and the formal apparatus of QM and GR (this paper) from the same postulates.

13 Relation to the Atomic Paper

Papers II and III rest on the same compact fiber from Paper I but address different questions. Paper II retains displacement information (the numbers (a, b, c) , screening slopes and intercepts, the quantum-defect law) that the Hilbert-space representation discards. Paper III derives the Hilbert representation itself, identifying what it preserves and what it loses. Together, they demonstrate that the compact fiber supports both the specific numerical content of atomic physics and the general formal structure of quantum mechanics.

14 Gravitation as Coordinate-Time Projection

Quantum mechanics is one of two effective descriptions that emerge from the RST postulates. The other is gravitation. Both arise from projections that discard structure, but the projections are different:

Effective theory	Projection	What is lost
Quantum mechanics	Covering map $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$	Kernel (vibrational phase)
General relativity	$3+3 \rightarrow 3+1$ (clock time only)	Coordinate time ($3v^2/c^2$)

This section demonstrates competency of the coordinate-time mechanism by reproducing the four classical tests of GR, deriving the PPN parameters $\beta = \gamma = 1$, establishing the equivalence principle, and identifying two testable second-order deviations from GR. The galactic-scale consequences of the gravitational limit and the coordinate-time mechanism—including the relationship between R^{16} and the observed acceleration scale of galaxy rotation curves—are the subject of a companion paper.

14.1 Coordinate-time mechanism

The three-dimensionality of time (P1 applied reciprocally via the reciprocal postulate) implies that the total time at any location comprises clock time (the uniform progression) plus three-dimensional *coordinate time*, analogous to three-dimensional coordinate space. Gravitation, being a three-dimensional scalar motion of speed v with

$$v^2 = \frac{GM}{r}, \quad (1)$$

produces a coordinate-time accumulation rate of $3v^2/c^2$ per unit of clock time—one factor of v^2/c^2 per scalar dimension [6].

$$\frac{dt_c}{dt} = \frac{3v^2}{c^2} = \frac{3GM}{rc^2}. \quad (2)$$

14.2 Equivalence principle

Proposition 14.1 (Weak equivalence principle). *The gravitational acceleration of a scalar-motion entity is independent of its displacement structure. Gravitational and inertial mass are identical.*

Proof. Step 1 (Inertial mass is total displacement content). By P1, motion exists in discrete units. The resistance of a scalar-motion entity to change in its state of motion is determined by its total displacement content: the number of occupied displacement steps across the three scalar dimensions. Mass has RST dimensions t^3/s^3 (one factor of t/s per scalar dimension). The total displacement content is specified by the entity's position on the compact fiber F , through the displacement triple (a, b, c) .

Step 2 (Gravitation acts on displacement content, not on a separate charge). Gravitation is three-dimensional scalar motion directed inward (Larson, point 34 of [3]). The scalar progression (P5) is universal outward motion at unit speed. Both are properties of motion itself, not of a coupling between motion and a separate field. The gravitational acceleration $a = GM/r^2$ depends on the source mass M and the distance r , but not on any property of the test entity.

Step 3 (Identity, not equality). The standard framework treats gravitational mass m_g and inertial mass m_i as two quantities that happen to be equal, confirmed experimentally to $\sim 10^{-15}$ (Eötvös-type experiments) but not explained by GR—it is an input. The RST does not contain two separate concepts: there is one quantity—total scalar displacement content—that determines both the entity's resistance to acceleration and the gravitational motion it sources. The equivalence principle is an identity of the displacement algebra, not a contingent equality.

Status: T. Step 1 from P1 and the RST definition of mass. Step 2 from the identification of gravitation as scalar motion. Step 3 is structural. $\square$

Remark 14.2 (Strong equivalence principle). *The strong form (local equivalence for all laws including self-gravitating systems) requires that gravitational binding energy contributes to inertial mass identically to other mass. In the RST, binding energy is displacement content of the same dimensional type (t^3/s^3). The strong principle follows from the same argument, with quantitative verification provided by the Nordtvedt parameter derived below ($\eta = 0$). Status: **S**.*

14.3 Reciprocal partition and the effective metric

Define the dimensionless gravitational potential

$$x \equiv \frac{GM}{rc^2}, \quad u \equiv \frac{x}{2} = \frac{GM}{2rc^2}. \quad (3)$$

The coordinate-time mechanism (Eq. 2) produces a total accumulation rate of $3x$ per unit clock time, distributed at x per scalar dimension. The fundamental RST postulate is that space and time are reciprocal aspects of motion. When gravitation produces coordinate time, the reciprocal postulate requires the effect to be partitioned symmetrically between its temporal and spatial manifestations:

- The temporal component is dilated by a factor $(1 - u)$: less proper time per unit coordinate time, because coordinate time absorbs part of the progression.
- The spatial component is contracted by a factor $(1 + u)$: more coordinate space per unit proper space, reciprocal to the temporal effect.

The factor $1/2$ (i.e., $u = x/2$) is forced by the reciprocal postulate: if the total gravitational effect is x and space and time enter symmetrically, the partition is $x/2$ to each. Any asymmetric partition would break the reciprocity.

The effective clock rate (proper time per coordinate time) is the ratio:

$$\frac{d\tau}{dt} = \frac{1 - u}{1 + u} = \frac{1 - x/2}{1 + x/2}. \quad (4)$$

The effective metric for a static, spherically symmetric mass is:

$$g_{00} = - \left(\frac{1 - u}{1 + u} \right)^2, \quad g_{rr} = 1 + 2\gamma x + O(x^2), \quad (5)$$

where g_{00} is derived from the reciprocal partition (status **T**) and the spatial metric is constrained to have $\gamma = 1$ at first order by the reciprocal postulate (status **S**). The second-order spatial structure depends on how the reciprocal postulate applies to the three-dimensional spatial metric and is not uniquely determined by the arguments of this section (status **O**).

Remark 14.3 (Correction to the Nehru redshift formula). *The gravitational redshift formula $f'/f = 1/(1+x)$ (Nehru [8]) is the first-order approximation: $1/(1+x) \approx 1-x$. The reciprocal-partition formula $f'/f = (1-u)/(1+u) = (1-x/2)/(1+x/2) \approx 1-x$ agrees at first order. They diverge at second order: $1/(1+x) = 1-x+x^2-\dots$ versus $(1-x/2)/(1+x/2) = 1-x+x^2/2-\dots$. The Nehru formula corresponds to $g_{00} = -1/(1+x)^2$, giving PPN parameter $\beta = 3/2$, which is excluded by lunar laser ranging ($|\eta| < 4.4 \times 10^{-4}$ requires $\beta = 1$). The reciprocal partition gives $\beta = 1$ and passes all current tests. The first-order results (all four classical tests below) are unchanged.*

14.4 PPN parameters

Expanding g_{00} from Eq. (5):

$$g_{00} = - \left(\frac{1-u}{1+u} \right)^2 = -(1 - 2x + 2x^2 - \frac{5}{2}x^3 + \dots). \quad (6)$$

Comparing with the PPN form $g_{00} = -(1 - 2x + 2\beta x^2 + \dots)$:

$$\boxed{\beta_{\text{RST}} = 1.} \quad (7)$$

The spatial metric at first order gives $g_{rr} = 1 + 2x + O(x^2)$, so:

$$\boxed{\gamma_{\text{RST}} = 1.} \quad (8)$$

These are identical to the GR values. The Nordtvedt parameter is $\eta = 4\beta - \gamma - 3 = 0$.

Status: $\beta = 1$ is **T** (from the reciprocal partition, Eq. 4). $\gamma = 1$ is **S** (from the reciprocal postulate applied to the spatial metric at first order).

14.5 Four classical tests

14.5.1 Perihelion precession

For a planet on an elliptical orbit with semi-latus rectum $l = a(1 - e^2)$, the coordinate-time accumulation over one revolution is [6]:

$$\delta = \frac{3GM}{lc^2} \times 2\pi = \frac{6\pi GM}{ac^2(1 - e^2)} \quad \text{radians/revolution.} \quad (9)$$

This is identical to the general-relativistic result. For Mercury: $\delta = 42.99''/\text{century}$, matching the observed $42.98 \pm 0.04''/\text{century}$.

The PPN combination $(2 + 2\gamma - \beta)/3 = 1$ confirms that the first-order precession formula is consistent with the full metric. The β correction enters at relative order $x \sim 10^{-8}$ for Mercury, far below measurement precision.

Status: **T**.

14.5.2 Gravitational deflection of light

For a photon grazing a mass M at closest approach R_0 , the coordinate-time accumulation produces a spatial displacement. Including the interaction cross-section for unpolarized radiation ($\sigma = 2/\pi$, averaged over all polarization orientations) [7]:

$$\delta = 2\pi \frac{v_0^2}{c^2} \times \frac{2}{\pi} = \frac{4GM}{R_0 c^2}. \quad (10)$$

This matches the general-relativistic prediction: $\delta = 1.751''$ for light grazing the Sun.

Status: **T**.

Remark 14.4 (Polarization-dependent deflection: a distinguishing prediction). *For polarized radiation, the interaction cross-section depends on the angle ϕ between the plane of polarization and the direction of gravity [7]:*

$$\delta_p = \frac{2\pi GM}{R_0 c^2} \cos \phi. \quad (11)$$

*GR predicts no polarization dependence. Quantitatively: for light grazing the Sun, the unpolarized deflection is $1.751''$. A fully polarized beam at $\phi = 0$ (polarization parallel to gravity) would be deflected by $2\pi GM/(R_0 c^2) = 2.750''$; at $\phi = 45^\circ$, by $1.945''$; at $\phi = 90^\circ$ (perpendicular), by $0''$. The difference between polarized ($\phi = 0$) and unpolarized deflection is $\sim 57\%$ —far above the current VLBI precision of $\sim 0.02\%$. This is a qualitative difference: GR predicts zero polarization dependence, the RST predicts a large $\cos \phi$ modulation. Status: **E**.*

14.5.3 Gravitational redshift

The effective clock rate from the reciprocal partition (Eq. 4) gives a frequency ratio:

$$\frac{f'}{f} = \frac{1-u}{1+u} = \frac{1-GM/(2rc^2)}{1+GM/(2rc^2)}. \quad (12)$$

At first order, $f'/f \approx 1 - GM/(rc^2)$, matching the GR result. The two predictions diverge at second order (see §14.7).

*Status: **T** at first order.*

14.5.4 Shapiro time delay

The coordinate speed of light in the RST effective metric is:

$$v_{\text{light}} = c \times \sqrt{\frac{-g_{00}}{g_{rr}}} \approx c(1-2x) \quad \text{at first order.} \quad (13)$$

The Shapiro delay for a signal passing near a mass M at closest approach d is:

$$\Delta t = \frac{(1+\gamma) \cdot 2GM}{c^3} \ln\left(\frac{4r_1 r_2}{d^2}\right) = \frac{4GM}{c^3} \ln\left(\frac{4r_1 r_2}{d^2}\right), \quad (14)$$

where r_1 and r_2 are the distances of emitter and receiver from the central mass, and $\gamma = 1$ has been used. This is identical to the GR prediction and is verified by the Cassini spacecraft measurement to relative precision $\sim 10^{-5}$ [20].

*Status: **T**.*

14.6 Observational summary: Solar System tests

Test	RST	GR	Precision
Perihelion precession	42.99"/century	42.98"/century	$\pm 0.04''$
Light deflection	$4GM/(R_0 c^2)$	$4GM/(R_0 c^2)$	$\sim 0.02\%$
Gravitational redshift	$1-x + O(x^2)$	$1-x + O(x^2)$	$\sim 10^{-4}$
Shapiro delay	$(4GM/c^3) \ln(\dots)$	$(4GM/c^3) \ln(\dots)$	$\sim 10^{-5}$
Cassini (γ)	$\gamma = 1$	$\gamma = 1$	$ \gamma - 1 < 2.3 \times 10^{-5}$
Nordtvedt (η)	$\eta = 0$	$\eta = 0$	$ \eta < 4.4 \times 10^{-4}$

All six Solar System tests are passed. The RST and GR predictions are identical through first order in $x = GM/(rc^2)$.

14.7 Second-order predictions: where RST and GR diverge

The RST and GR effective metrics agree at first order but diverge at second order. Two predictions are cleanly falsifiable.

14.7.1 Second-order gravitational redshift

$$\text{RST: } f'/f = \frac{1-x/2}{1+x/2} = 1-x + \frac{1}{2}x^2 - \frac{1}{4}x^3 + \dots, \quad (15)$$

$$\text{GR: } f'/f = \sqrt{1-2x} = 1-x - \frac{1}{2}x^2 - \frac{1}{2}x^3 - \dots. \quad (16)$$

The second-order coefficients have opposite signs: $+\frac{1}{2}x^2$ (RST) versus $-\frac{1}{2}x^2$ (GR). The relative difference scales as x^2 :

System	$x = GM/(rc^2)$	Relative diff.	Detectability
Sun	2×10^{-6}	$\sim 10^{-12}$	Not detectable
Sirius B	2.6×10^{-4}	$\sim 10^{-4}$	Marginal
Neutron star	~ 0.2	$\sim 5.6\%$	Detectable

For a neutron star with $x \approx 0.2$: RST predicts $f'/f = 0.818$; GR predicts $f'/f = 0.775$. X-ray spectroscopy of redshifted absorption lines from neutron star surfaces (current precision $\sim 5\%$ for some sources) provides a viable test. *Status: E*.

14.7.2 Polarization-dependent deflection

As noted in Remark 14.4, the RST predicts a $\cos \phi$ dependence of light deflection on polarization angle, while GR predicts none. A single VLBI measurement comparing deflection of polarized radio sources at different polarization angles near the Sun would falsify one theory or the other. *Status: E*.

14.8 Gravitational wave speed

Changes in the gravitational configuration propagate through the scalar progression, which operates at unit speed c (P5). The speed of gravitational disturbances is therefore $v_{\text{gw}} = c$, consistent with the LIGO/Virgo measurement $|v_{\text{gw}}/c - 1| < 10^{-15}$ from the binary neutron star merger GW170817/GRB 170817A [21]. The polarization structure and quadrupole radiation formula require treatment of the tensorial content of the coordinate-time field and are deferred.

Status: Speed is **S** (structural consequence of P5). Polarization and radiation formula are **O**.

14.9 The gravitational limit

The RST postulates predict a characteristic feature absent from general relativity: the *gravitational limit* (Larson, point 37 of [3]). Because the scalar progression is always outward at unit speed and gravitation is inward with magnitude decreasing as r^{-2} , the two motions reach equality at some distance r_{GL} . Beyond this distance, the net motion is outward. The gravitational limit is a function of the central mass; its quantitative consequences for galactic dynamics are a target for the companion paper on rotation curves.

Status: **S** for the existence of the limit (structural consequence of the progression and gravitational speed profiles). **O** for the quantitative application to rotation curves.

14.10 The hierarchy ratio

The RST framework does not have independent coupling constants that require fine-tuning.

(i) *Forces are not fundamental.* Gravitation is three-dimensional scalar motion inward (Larson, point 34). Electric charge is one-dimensional rotational vibration (points 114–117). These are different kinds of motion, not different instances of the same kind, and their “strengths” are not comparable in the way that $\alpha/\alpha_G \approx 10^{36}$ suggests.

(ii) *Physical constants are not independent.* In the RST, there are only two independent quantities: the natural unit of space ($s_{\text{nat}} = 4.559 \times 10^{-8}$ m) and the natural unit of time ($t_{\text{nat}} = 1.521 \times 10^{-16}$ s). All physical constants are derived from these plus the inter-regional ratio $R = 156.444$ and the fiber structure.

(iii) *The power R^{16} appears in the dimensional analysis.* The inter-regional ratio raised to the sixteenth power, $R^{16} \approx 1.29 \times 10^{35}$, arises naturally when connecting the natural acceleration scale $a_{\text{nat}} = s_{\text{nat}}/t_{\text{nat}}^2$ to conventional gravitational scales through the dimensional structure of mass (t^3/s^3). This is suggestive of the observed electromagnetic-to-gravitational coupling ratio

$\alpha/\alpha_G \approx 10^{36}$, but the exact relationship requires deriving G from the natural unit system, which has not been completed.

Status: O. The structural ingredients are in place (derived constants, dimensional analysis, the R^{16} scale), but closing the hierarchy argument requires the G derivation. The quantitative relationship between R^{16} and the galactic acceleration scale is developed in the companion paper on rotation curves.

14.11 What the gravitational section does not claim

The gravitational program is staged. This section demonstrates that the coordinate-time mechanism reproduces all four classical tests and all current precision Solar System constraints ($\beta = \gamma = 1$, $\eta = 0$). It does not derive the Einstein field equations, does not treat gravitational wave polarization or radiation, and does not address the strong-field regime. These are identified as open formalization tasks.

Two predictions are cleanly falsifiable with current or near-term technology:

1. **Polarization-dependent deflection.** A VLBI measurement of polarized vs. unpolarized radio source deflection near the Sun at current precision ($\sim 0.02\%$) would confirm or rule out the $\cos \phi$ dependence.
2. **Second-order gravitational redshift.** X-ray spectroscopy of neutron star surfaces could distinguish the RST prediction $f'/f = (1 - x/2)/(1 + x/2)$ from the GR prediction $f'/f = \sqrt{1 - 2x}$ at the $\sim 5\%$ level for $x \approx 0.2$.

Either measurement returning the GR prediction would falsify the coordinate-time mechanism as the origin of these effects.

14.12 Structural parallel with quantum mechanics

The two effective descriptions share a common logical structure:

1. Both arise from projections that discard temporal structure.
2. Both reproduce the standard theory's predictions within their first-order domains.
3. Both produce testable deviations where the projection breaks down or where second-order effects become accessible.
4. Both are derived from the same postulates without importing the structures they explain.

The quantum-mechanical projection (through π) discards the covering kernel and produces the Born-rule probability structure. The gravitational projection (from 3+3 to 3+1) discards coordinate time and produces the effective metric structure. Neither projection is more fundamental than the other; both are consequences of the observer's anchorage in the three-dimensional spatial reference frame.

14.13 Gravitational program status

Claim	Status	Depends on / Note
Newtonian limit ($v^2 = GM/r$)	T	3D scalar motion inward
Equivalence principle	T	Identity of gravitational and inertial mass (Prop. 14.1)
Perihelion precession ($42.99''$)	T	Coordinate-time rate $3v^2/c^2$
Light deflection ($4GM/R_0c^2$)	T	Coordinate-time + cross-section
Gravitational redshift (1st order)	T	Reciprocal partition
PPN $\beta = 1$	T	Reciprocal partition of gravitational potential
Shapiro time delay	T	$\gamma = 1 +$ coordinate light speed
Nordtvedt $\eta = 0$	T	$\beta = \gamma = 1$
PPN $\gamma = 1$ (1st order)	S	Reciprocal postulate on spatial metric
Strong equivalence principle	S	Same displacement-content argument + $\eta = 0$
GW speed = c	S	Progression speed (P5)
Coordinate-time mechanism	I	Identification of $3+3 \rightarrow 3+1$ projection
Second-order redshift	E	$+x^2/2$ (RST) vs. $-x^2/2$ (GR); testable at neutron stars
Polarization-dependent deflection	E	$\cos \phi$ (RST) vs. none (GR); testable by VLBI
Effective field equations	O	Metric not yet promoted to field equation
GW polarization and radiation	O	Tensorial content of coordinate-time field
Strong-field regime	O	Non-perturbative coordinate-time
Spatial metric (2nd order)	O	Reciprocal postulate on 3D spatial metric
G from natural units	O	Required to close hierarchy argument
Gravitational limit $\rightarrow$ rotation curves	O	Deferred to companion paper

15 Conclusion

The main result is:

Effective quantum formalism is induced by the compact fiber and cross-frame projection. Under stated assumptions (identified in the derivation-status ledger, §1.5), Hilbert space, the Born rule, complex amplitudes, superposition, entanglement, Bell correlations, the measurement-problem resolution, and preferred-basis selection emerge as representational consequences of the compact stable local carrier.

The derivation chains, with status labels:

Quantum program (covering-map projection $\pi : \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$):

SELECTOR GEOMETRY (bundle, branches, carrier)	
+ bilinear integration + conservation + continuity	
↓	
LINEAR EXPANSION $\psi = \sum c_i B_i$	[T, Lem. 6.1]
↓	
L^2 NORM $\sum c_i^2 = 1$	[T, Lem. 6.2]
↓	

INNER PRODUCT $\langle i \psi\rangle = c_i$	[T , Cor. 6.3]
↓	
WEIGHT RULE $W_i = c_i^2$	[T]
↓	
CG2: COMPLEXIFICATION	[T , Lem. 6.4]
+ phase gauge invariance	[T , Lem. 6.5]
↓	
BORN RULE $P_i = c_i ^2$	[S/T , Thm. 6.6]
↓	
CG3: TENSOR PRODUCT	[S]
↓	
ENTANGLEMENT ($\theta > 0$)	[S]
↓	
DECOHERENCE ($\eta < 1$)	[T]
↓	
CLASSICAL SELECTOR WEIGHTS	[T]

Gravitational program (coordinate-time projection $3+3 \rightarrow 3+1$):

COORDINATE-TIME MECHANISM ($3+3 \rightarrow 3+1$ projection)	
+ reciprocal postulate (space-time symmetry)	
↓	
EQUIVALENCE PRINCIPLE	[T , Prop. 14.1]
↓	
RECIPROCAL PARTITION $d\tau/dt = (1-u)/(1+u)$	[T , Eq. 4]
↓	
PPN $\beta = 1, \gamma = 1$	[T/S , §14.4]
↓	
FOUR CLASSICAL TESTS (precession, deflection, redshift, Shapiro)	[T]
↓	
SECOND-ORDER PREDICTIONS (redshift sign, polarization)	[E]

The status labels reflect the current state of the derivation program. The two items that were status **O** in the previous draft—cross-frame factorization and antisymmetric state selection—have been upgraded to **T** by deriving displacement additivity from the postulates and the singlet state from P5 via $J_{\text{total}} = 0$. Both upgrades produce testable predictions confirmed by experiment. All quantum claims are now **T** except the single interpretive dissolution step (Proposition 8.2); gravitational claims are **T** through first order, with two second-order predictions at status **E** (testable, not yet tested).

15.1 Proof-status summary

Claim	Status	Depends on
Complexification	T	Covering fiber + quarter-turn uniqueness (Prop. 4.2)
Inner product	T	Bilinearity + conservation + continuity (Lem. 6.2)
Born rule	S/T	L^2 uniqueness + phase gauge + pre-CG2 weight
Projection/collapse	T	Displacement additivity (Lem. 5.3) + π -uniqueness
Preferred basis	T	Detector coupling diagonalization (Prop. 8.1)
Bell correlations	T	P5 datum neutrality $\rightarrow$ singlet (Lem. 9.1)
Pauli exclusion	T	Kernel action (Lem. 5.1) + rotational rep
Equivalence principle	T	Identity via displacement content (Prop. 14.1)
PPN $\beta = \gamma = 1$	T/S	Reciprocal partition (§14.4); β : T , γ (1st order): S
Shapiro delay	T	$\gamma = 1$
GW speed = c	S	Progression speed (P5)

16 Open Formalization Tasks

1. **Time-region dynamics for the kernel component.** The structural dissolution of the measurement problem (§8.5) identifies the kernel component of the fiber state as the information that determines individual outcomes. Whether this component evolves deterministically or stochastically at the fiber level is a question about the completeness of the time-region mechanics, not about the measurement process. Resolving this would either (a) make individual-outcome prediction possible in principle (if deterministic), or (b) identify the fiber level as the deepest level at which irreducible randomness enters (if stochastic).
2. **Continuous observables.** The identification of $L^2(\mathbb{R}^+)$ as the time-region representation space needs formal boundary-condition treatment at $\rho = 0$.
3. **Decoherence dynamics.** A quantitative criterion for “acquisition of independent motion” from the displacement algebra, beyond the phenomenological $\eta = T_2/T_{\text{osc}}$ criterion.
4. **Quaternion extension.** Nehru [5] shows that the three-dimensional zone of the time region (2D rotation, the magnetic sector) requires quaternion wave functions $\psi = \{\psi_a, i\psi_c, j\psi_b, k\psi_d\}$ rather than complex ones. The relationship between this quaternion structure and the covering-fiber geometry of Paper I has not yet been formalized. This may bear on the extension of the present framework to baryon and nuclear structure.
5. **Spatial metric second-order structure.** The reciprocal partition (§14.3) uniquely determines the temporal metric g_{00} and the first-order spatial metric ($\gamma = 1$). The second-order spatial metric depends on how the reciprocal postulate applies to the three-dimensional spatial sector, which involves a subtlety (spatial is inherently 3D, temporal is 1D) not resolved by the arguments of §14.4. The simplest reciprocal identification gives $g_{rr} = ((1+u)/(1-u))^2$, which differs from the GR isotropic form $(1+u)^4$ at order u^2 . The difference is undetectable at Solar System scales ($u \sim 10^{-8}$) but predicts $\sim 1.4\%$ deviation in spatial geometry at neutron star surfaces ($u \sim 0.1$).

6. **Effective field equations.** The RST effective metric (§14.3) passes all PPN tests, but the field equation it satisfies has not been derived. The temporal component $g_{00} = -((1-u)/(1+u))^2$ and the spatial structure (once the second-order form is determined) define a specific metric; computing the Ricci tensor of this metric and identifying the corresponding field equation is a mechanical but non-trivial task. Whether the result is the Einstein field equation (in which case the RST provides an alternative derivation of GR) or a modified field equation (in which case the RST makes additional predictions beyond the PPN level) is an open question.
7. **Gravitational wave polarization and radiation.** The coordinate-time mechanism predicts gravitational wave speed $= c$ (§14.8), consistent with LIGO/Virgo observations. The number of independent polarization modes and the quadrupole radiation formula require extending the scalar-motion formalism to describe time-varying mass distributions and their coordinate-time fields.

A Bell/CHSH Computation Details

A.1 Photon polarization

The joint state is $|\Psi\rangle = (|s_1 s_2\rangle - |s_2 s_1\rangle)/\sqrt{2}$.

Alice measures at angle 0, Bob at angle θ . Computing the amplitude $A(+, +)$:

$$\begin{aligned} A(+, +) &= (\langle s_1 | \otimes \langle m_+^B(\theta) |) |\Psi\rangle \\ &= \frac{1}{\sqrt{2}} [\langle s_1 | s_1 \rangle \langle m_+^B | s_2 \rangle - \langle s_1 | s_2 \rangle \langle m_+^B | s_1 \rangle] \\ &= \frac{1}{\sqrt{2}} [1 \cdot \sin \theta - 0] = \frac{\sin \theta}{\sqrt{2}}. \end{aligned}$$

Similarly: $A(+, -) = \cos \theta / \sqrt{2}$, $A(-, +) = -\cos \theta / \sqrt{2}$, $A(-, -) = \sin \theta / \sqrt{2}$.

CHSH at optimal angles: $S = E(\pi/8) + E(3\pi/8) + E(\pi/8) - E(-\pi/8) = 2\sqrt{2}$.

A.2 Classical hidden-variable bound

For hidden variable λ (shared polarization angle):

$$E_{\text{mix}}(\theta) = \int_0^\pi \frac{1}{\pi} [2 \cos^2 \lambda - 1] [2 \sin^2(\lambda - \theta) - 1] d\lambda = -\frac{1}{2} \cos 2\theta.$$

$$|S_{\text{mix}}| = \sqrt{2} \leq 2.$$

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